

CENG381 Stochastic Processes

Classifying States - Answer Key 1

Q1 Solution

a) Transition diagram:

0 --[1.0]--> 1

1 --[0.5]--> 0, 1 --[0.5]--> 2

2 --[0.5]--> 1, 2 --[0.5]--> 3

3 --[0.4]--> 3 (self-loop), 3 --[0.6]--> 4

4 --[1.0]--> 4 (self-loop, absorbing)

b) Communicating classes:

Can 0 reach 1? Yes (direct in 1 step).

Can 1 reach 0? Yes (1 -> 0 in 1 step).

Can 1 reach 2? Yes. Can 2 reach 1? Yes.

=> States 0, 1, 2 all communicate: Class A = {0, 1, 2}

Can 2 reach 3? Yes (2 -> 3). Can 3 reach 2? No (3 only goes to 3 or 4).

=> State 3 does NOT communicate with {0,1,2}.

Can 3 reach 4? Yes. Can 4 reach 3? No (4 is absorbing).

=> Class B = {3} (singleton).

Class C = {4} (absorbing, singleton).

Summary: Class A = {0,1,2}, Class B = {3}, Class C = {4}.

c) Classification of each class:

Class A = {0, 1, 2}:

States 0, 1, 2 can reach state 3 (via 2 -> 3), and once in 3 they can never return to {0,1,2}. So there is a positive probability of leaving and never returning. => Class A is TRANSIENT.

Class B = {3}:

State 3 can move to state 4 (prob 0.6), which is absorbing. So from state 3 there is a positive probability (0.6) of leaving and never returning. => Class B is TRANSIENT.

Class C = {4}:

State 4 is absorbing: $P(4,4) = 1$, it never leaves.

Once entered, the chain stays forever. => Class C is POSITIVE RECURRENT

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(and absorbing). Limiting probability: $\pi(4) = 1$.

Q2 Solution

a) Transition matrix P (order: R, D, U, S):

	R	D	U	S
R	[0.8	0.2	0.0	0.0]
D	[0.0	0.5	0.3	0.2]
U	[0.7	0.0	0.3	0.0]
S	[0.0	0.0	0.0	1.0]

b) Communicating classes:

R $\rightarrow$ D (prob 0.2), D $\rightarrow$ U (prob 0.3), U $\rightarrow$ R (prob 0.7).

So R, D, U all communicate with each other: Class A = {R, D, U}.

S is absorbing and cannot reach R, D, or U: Class B = {S}.

Class A = {R, D, U}: From D, there is a path to S (prob 0.2 direct).

Once in S, the chain cannot return to A. $\Rightarrow$ Class A is TRANSIENT.

Class B = {S}: Absorbing state. $\Rightarrow$ Class B is POSITIVE RECURRENT.

c) Absorption probability into S starting from each state:

Let $h(i) = P(\text{eventually reach S} \mid \text{start in state } i)$.

Clearly $h(S) = 1$.

Balance equations (condition on first step):

$$\begin{aligned} h(R) &= P(R,R)h(R) + P(R,D)h(D) + P(R,U)h(U) + P(R,S)h(S) \\ &= 0.8h(R) + 0.2h(D) + 0.0h(U) + 0.0 \end{aligned}$$

$$\begin{aligned} h(D) &= P(D,D)h(D) + P(D,U)h(U) + P(D,S)h(S) \\ &= 0.5h(D) + 0.3h(U) + 0.2 \end{aligned}$$

$$\begin{aligned} h(U) &= P(U,R)h(R) + P(U,U)h(U) \\ &= 0.7h(R) + 0.3h(U) \end{aligned}$$

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Simplify each equation:

$$\text{From } h(R): 0.2 \cdot h(R) = 0.2 \cdot h(D) \Rightarrow h(R) = h(D) \quad \dots (i)$$

$$\text{From } h(D): 0.5 \cdot h(D) = 0.3 \cdot h(U) + 0.2 \Rightarrow h(D) = 0.6 \cdot h(U) + 0.4 \quad \dots (ii)$$

$$\text{From } h(U): 0.7 \cdot h(U) = 0.7 \cdot h(R) \Rightarrow h(U) = h(R) \quad \dots (iii)$$

From (i) and (iii): $h(R) = h(D) = h(U) = h$ (all equal).

$$\text{Substituting into (ii): } h = 0.6 \cdot h + 0.4 \Rightarrow 0.4 \cdot h = 0.4 \Rightarrow h = 1.$$

Result: $h(R) = h(D) = h(U) = 1$.

The machine is GUARANTEED to eventually be scrapped, regardless of starting state (since every class A state is transient and S is the only recurrent state).

Q3 Solution

a) Communicating classes (from transition matrix):

States 0 and 1:

$0 \rightarrow 1$ (prob $2/3 > 0$), $1 \rightarrow 0$ (prob $1/3 > 0$) $\Rightarrow$ 0 and 1 communicate.

Neither 0 nor 1 can reach 2, 3, 4, or 5. $\Rightarrow$ Class A = $\{0, 1\}$.

State 2:

$2 \rightarrow 1$ (prob $3/5$) but 1 cannot reach 2. So 2 does NOT communicate with class A. $2 \rightarrow 3$ (prob $1/5$), 3 cannot reach 2. $\Rightarrow$ Class B = $\{2\}$.

States 3 and 4:

$3 \rightarrow 4$ (prob $3/4$), $4 \rightarrow 3$ (prob $1/2$) $\Rightarrow$ 3 and 4 communicate.

Neither can reach 0, 1, 2, or 5. $\Rightarrow$ Class C = $\{3, 4\}$.

State 5:

$P(5,5) = 1$, absorbing. $\Rightarrow$ Class D = $\{5\}$.

b) Classification:

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Class A = {0, 1}: A is a closed class (no transitions leave A).

It is finite, irreducible, so it is POSITIVE RECURRENT.

Class B = {2}: State 2 can reach class A (via 2→1) and class C (via 2→3). It can never return once it leaves 2 (since A and C are closed or lead to D). => Class B is TRANSIENT.

Class C = {3, 4}: It is finite, irreducible, and closed (3 and 4 only transition within {3,4}). => Class C is POSITIVE RECURRENT.

Class D = {5}: Absorbing. => POSITIVE RECURRENT.

c) Limiting state probabilities:

Transient states: $\pi(2) = 0$ in the limit.

The chain starting from any state will eventually be absorbed into class A, class C, or class D, depending on the starting state.

Since the question asks for limiting probabilities without specifying a starting distribution, we find the stationary distribution within each recurrent class:

Within Class A = {0, 1}: solve $\pi_A * P_A = \pi_A$

$$P_A = \begin{bmatrix} 1/3 & 2/3 \\ 1/3 & 2/3 \end{bmatrix}$$

$$\pi_A(0) \cdot (1/3) + \pi_A(1) \cdot (1/3) = \pi_A(0)$$

$$\pi_A(0) + \pi_A(1) = 1$$

$$\text{From row 0: } (2/3) \cdot \pi_A(0) = (1/3) \cdot \pi_A(1) \Rightarrow \pi_A(1) = 2 \cdot \pi_A(0)$$

$$\Rightarrow \pi_A(0) = 1/3, \pi_A(1) = 2/3$$

Within Class C = {3, 4}: solve $\pi_C * P_C = \pi_C$

$$P_C = \begin{bmatrix} 1/4 & 3/4 \\ 1/2 & 1/2 \end{bmatrix}$$

$$\pi_C(3) = (1/4) \cdot \pi_C(3) + (1/2) \cdot \pi_C(4)$$

$$\pi_C(3) + \pi_C(4) = 1$$

$$(3/4) \cdot \pi_C(3) = (1/2) \cdot \pi_C(4) \Rightarrow \pi_C(4) = (3/2) \cdot \pi_C(3)$$

$$\Rightarrow \pi_C(3) = 2/5, \pi_C(4) = 3/5$$

Class D = {5}: $\pi_D(5) = 1$.

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The overall long-run distribution depends on the starting state.

Starting from state 2 (transient), the chain is eventually absorbed into A (with some probability α) or C or D (with prob $1-\alpha$).